

APEX OPTIONS INTELLIGENCE PLATFORM

AI-Driven Options Trading Intelligence — Full Product Pitch, User Flow & Development Guide

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EXECUTIVE SUMMARY

APEX is a professional-grade, fully autonomous AI options trading platform. A trader types any U.S. equity ticker, selects a chart timeframe and options expiry from a dropdown, hits SCAN — and within 15 seconds the platform has:

1. Run a 10-layer technical analysis on the price chart (MACD, RSI, EMA stack, SuperTrend, Bollinger Bands, Candlestick Patterns, Pivot Points, Support/Resistance)
2. Loaded the live options chain and filtered every strike by Delta, Theta, Vega, Gamma, and bid/ask spread
3. Compared Implied Volatility (IV) to Historical Volatility (HV) to determine whether to buy or sell premium
4. Processed real-time sentiment from financial news and options flow
5. Pulled a fundamental snapshot: earnings history, EPS trend, analyst consensus, revenue growth
6. Synthesized all five layers into a single **APEX Composite Score (0–100)**
7. Selected the optimal strategy from its approved playbook (Iron Condor, Bull Call Spread, Bear Put Spread, Gamma Trampoline™, and more)
8. Presented a full trade brief in plain English — and executed autonomously when the score clears the 72/100 threshold

APEX is not a signal dashboard. It is a **full-cycle AI trading co-pilot** that thinks like an expert options desk, executes with institutional discipline, and explains every decision in plain language.

The platform's **proprietary innovation** is the **Gamma Trampoline™** — an original options strategy that exploits IV inversion around catalyst events (earnings, product launches) to dramatically reduce the cost of a standard straddle while maintaining full long-gamma upside.

Market Problem: Options trading at the professional level requires simultaneous evaluation of dozens of variables across multiple platforms. APEX collapses all of that into one 15-second

scan and one decisive recommendation.

Technology Stack: Python (FastAPI backend) + React frontend + multi-model AI consensus (GPT-4, Claude, Gemini) + live market data APIs + NLP sentiment pipeline.

Revenue Model: Subscription SaaS targeting professional traders, proprietary trading desks, and financial advisory firms.

SECTION 1 — THE PROBLEM WE ARE SOLVING

Modern options trading demands the simultaneous evaluation of dozens of variables: price action, momentum, volatility structure, time decay, directional bias, liquidity, sentiment, and fundamental context. Even experienced traders suffer from:

- **Analysis Paralysis** — Too many indicators across too many tabs, with no convergent signal
- **Volatility Mispricing** — Entering long positions when IV is already inflated, or selling when IV is historically cheap
- **Greek Blindness** — Opening positions without quantifying Theta drag, Gamma acceleration, or Vega exposure
- **Execution Inconsistency** — Excellent individual analysis but no structured, repeatable framework
- **Strategy Mismatch** — Applying income strategies in trending markets, or directional strategies in flat environments

APEX solves all five — simultaneously, in under 15 seconds, on any publicly traded equity.

SECTION 2 — PRODUCT VISION & THE THINK 10X FRAMEWORK

Vision Statement

"Every trade deserves institutional-grade analysis. APEX delivers that — automatically, transparently, and decisively."

The THINK 10X Engine

No single metric fires the trigger. All ten dimensions vote. The AI synthesizes consensus and acts.

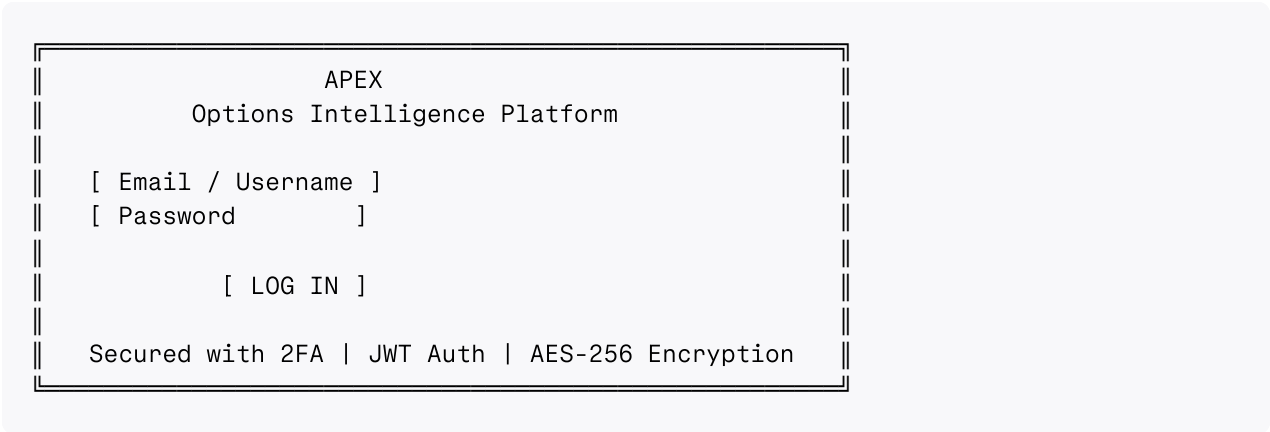
Dimension	What APEX Measures
1. Trend	SuperTrend + EMA Stack (9/21/50/100/200) — is price in a confirmed trend?
2. Momentum	RSI (14) overbought/oversold + MACD crossovers and histogram divergences
3. Volatility	Bollinger Band width + IV vs. HV comparison + IVR (Implied Volatility Rank)
4. Structure	Candlestick pattern recognition — engulfing, doji, hammer, shooting star, morning/evening star

Dimension	What APEX Measures
5. Levels	Pivot Points (PP, R1–R3, S1–S3) + dynamic support/resistance from volume clusters
6. Greeks	Delta, Theta, Vega, Gamma — option-level fitness scoring per strike
7. Chain	Options chain — OI, volume, bid/ask spread, put/call ratio, unusual activity alerts
8. Sentiment	News NLP + social media scoring + options flow direction
9. Fundamentals	P/E, EPS trend, earnings calendar, revenue growth, analyst consensus
10. Strategy	Final synthesis layer — selects and executes optimal strategy from the approved playbook

SECTION 3 — FULL TERMINAL USER EXPERIENCE (UX WALKTHROUGH)

3.1 Application Launch & Login Screen

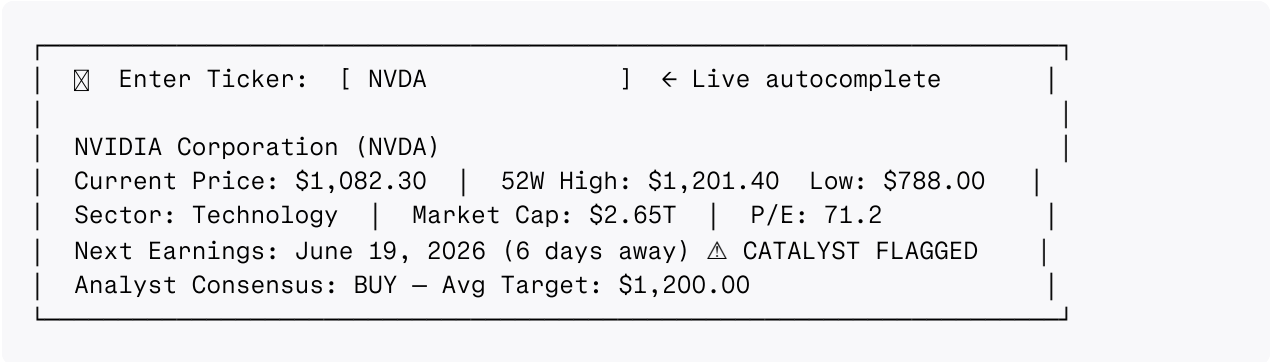
User opens APEX in the browser (or desktop app wrapper).



- After login, the user lands on the **Main Dashboard**
- Dashboard shows: watchlist tickers, market sentiment indicator (bullish/bearish/neutral), and any open positions being monitored

3.2 Step 1 — Ticker Entry

User types a ticker into the search bar at the top of the screen.



What the system does automatically at this step:

- Pulls company name, logo, current OHLCV price data
- Flags upcoming earnings or known catalyst dates in orange
- Displays a pre-load fundamental card
- Populates the expiry dropdown with available options expirations

3.3 Step 2 — Select Timeframe & Expiry Date

Two dropdowns appear alongside the ticker.

Timeframe: [Daily (1D) ▼] Expiry: [Jun 20, 2026 (7d) ▼]

Timeframe Options:

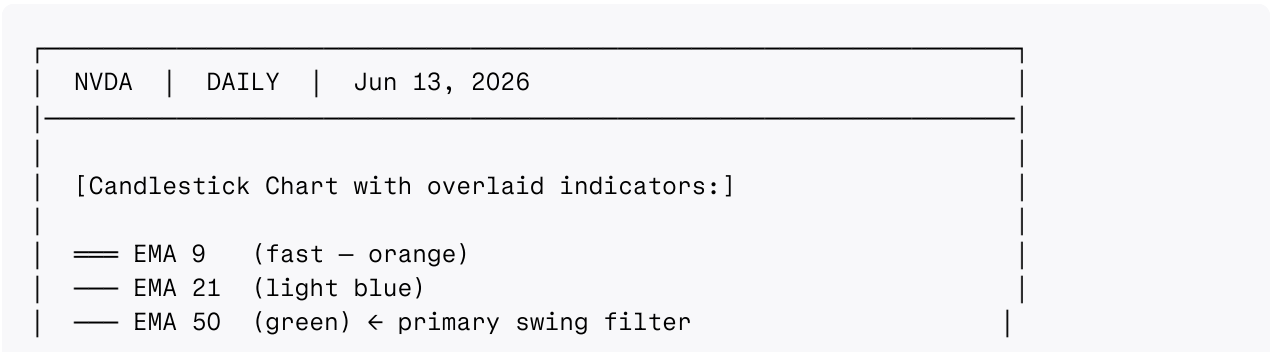
Code	Label	Best Use
1m	1 Minute	Scalping / Day trading
5m	5 Minutes	Active intraday
15m	15 Minutes	Momentum intraday
1H	1 Hour	Swing confirmation
4H	4 Hours	Multi-day positioning
1D	Daily	Standard swing / weekly options
1W	Weekly	LEAPS / multi-month positions

Expiry Options (pulled live from options chain API):

- Sorted chronologically: current week → next week → monthly → quarterly → LEAPS
- Upcoming earnings dates highlighted in amber
- Near-expiry (within 2 days) highlighted in red

3.4 Step 3 — The Chart Loads

The interactive candlestick chart renders immediately with all indicators overlaid.



- EMA 100 (purple)
- EMA 200 (red) ← major trend gate
- █ SuperTrend line (green below price = bullish)
- ... Bollinger Bands (upper / mid / lower)
- - Pivot Levels (PP, R1, R2, S1, S2 as dashed lines)
- ▒ Support/Resistance zones (shaded horizontal bands)

[Sub-panel 1: MACD histogram + signal line]

[Sub-panel 2: RSI line with 70/30 reference bands]

All indicators are toggle-able via a sidebar. Defaults are all ON.

3.5 Step 4 — User Clicks [SCAN]

This is the core event that triggers the entire APEX analysis engine.

A progress panel appears on the right side of the screen, updating in real-time:

APEX SCANNING: NVDA	
✓ Technical Analysis	Analyzing... DONE
✓ Candlestick Patterns	Bullish Engulfing
✓ MACD	Bullish Cross ↑
✓ RSI (14)	51 — Neutral
✓ EMA Stack	Aligned Bullish
✓ SuperTrend	Bullish (Price Above)
✓ Bollinger Bands	Neutral Width
✓ Pivot Points	Near R1 — \$1,095
✓ Support / Resistance	Strong S at \$1,040
✓ Options Chain Loaded	Apr 20 — 248 Strikes
✓ Greeks Analysis	Filtered to 14 Strikes
✓ Volatility Engine	IV: 72% HV: 38%
↳ IV >> HV — SELL PREMIUM SIGNAL	
↳ IVR: 78 — Rich Premium Environment	
↳ Front-Week IV > Back-Week: IV INVERSION ACTIVE	
✓ Sentiment Analysis	Bullish (Score: 68/100)
✓ Fundamental Score	Strong (Score: 74/100)
⚡ APEX COMPOSITE SCORE:	84 / 100 ✓
⚡ THRESHOLD MET — STRATEGY SELECTED	

3.6 Step 5 — The APEX Trade Recommendation Card

After the scan completes, the recommendation card slides into view:

APEX TRADE EXECUTION BRIEF		
Ticker:	NVDA	Timeframe: 1D Score: 84 / 100
Strategy:	GAMMA TRAMPOLINE™	
Catalyst:	EARNINGS – 6 days away ▲	
THE FOUR LEGS:		
LEG 1 (Long):	Buy NVDA Jun27 \$1,100 Call	@ \$18.40
LEG 2 (Long):	Buy NVDA Jun27 \$1,040 Put	@ \$16.20
LEG 3 (Short):	Sell NVDA Jun20 \$1,100 Call	@ \$11.80
LEG 4 (Short):	Sell NVDA Jun20 \$1,040 Put	@ \$9.90
NET DEBIT: \$12.90 per spread		
MAX LOSS:	\$12.90	TARGET PROFIT: \$38.00+
P.O.P (AI Estimate):	54%	
WHY THIS STRATEGY:		
"NVDA earnings in 6 days. Front-week IV (72%) significantly exceeds back-week IV (38%) – IV inversion confirmed. RSI is neutral at 51 with MACD bullish cross. Three consecutive EPS beats with revenue +18% YoY. Gamma Trampoline selected to exploit IV inversion: selling front-week volatility subsidizes the back-week long options by ~65%. All legs pass bid/ask filter (< 8% of mid). Net Delta ≈ zero. APEX Score: 84."		
GREEKS SUMMARY:		
Net Delta:	-0.02 (near neutral)	Net Theta: +\$4.20/day
Net Vega:	+0.32	Net Gamma: +0.08
BENCHMARK CHECKS:		
✓ IV > HV by 34pts – Sell premium confirmed		
✓ Front-week Delta < 0.25 on short legs		
✓ Bid/Ask spread < 8% on all four legs		
✓ OI > 500 on all strikes		
✓ No earnings within front-week expiry window		
[MODIFY PARAMETERS] [EXECUTE TRADE]		

3.7 Step 6 — Post-Execution Monitoring Dashboard

After a trade executes, the **Position Monitor** activates automatically:

ACTIVE POSITION: NVDA — GAMMA TRAMPOLINE™		
Opened:	Jun 13, 2026	DTE: 7 days (front) / 14 (back)
Net Debit:	\$12.90	Current P&L: +\$3.40 (+26.4%)

LIVE GREEK MONITOR:

Delta: -0.01 ✓ | Theta: +\$4.80/day ✓ | Vega: +0.29 ✓

ALERTS ACTIVE:

☒ Profit Target (40% of max): \$15.48 – not yet reached

☒ Stop Loss (100% of debit): -\$12.90 – not triggered

☒ Earnings Day: Jun 19 – APEX monitoring for IV crush

☒ Bid/Ask Health: All legs within 8% spread ✓

Automated exit rules (AI-managed):

- Take profit at 40% of max gain (default)
- Cut loss at 100% of net debit (default)
- Close 1 day before expiry if still open
- Alert if IV crushes post-earnings and Vega exposure flips

SECTION 4 — DETAILED TECHNICAL ANALYSIS ENGINE

4.1 Candlestick Pattern Recognition

The AI computer vision module scans the most recent candle formations:

Signal Type	Pattern	Meaning
Bullish Reversal	Hammer, Bullish Engulfing, Morning Star, Dragonfly Doji	Buy signal at support
Bearish Reversal	Shooting Star, Bearish Engulfing, Evening Star, Dark Cloud Cover	Sell signal at resistance
Continuation	Three White Soldiers, Three Black Crows	Trend continuation
Indecision	Spinning Top, Standard Doji, Inside Bar	Requires confirmation

4.2 MACD Signals

Standard configuration: 12-period EMA, 26-period EMA, 9-period signal line.

- **Bullish:** MACD crosses above signal; histogram turning positive; MACD rising from below zero
- **Bearish:** MACD crosses below signal; histogram expanding negatively
- **Divergence:** Price makes new high but MACD does not → bearish divergence (and vice versa) — flagged as high-confidence reversal

4.3 RSI (14-period)

- RSI > 70: Overbought → favor puts or bearish spreads
- RSI < 30: Oversold → favor calls or bullish spreads
- RSI 40–60 (neutral): Range-bound → favor iron condors or premium selling
- Divergence between RSI and price → reversal signal

4.4 EMA Stack Logic

EMA	Period	Role
Fast	9	Short-term momentum
Short	21	Trend confirmation for weekly options
Mid	50	Primary swing trend filter
Key	100	Institutional support/resistance
Long	200	Major trend gate; SuperTrend baseline

EMA Stacking Rule: When $9 > 21 > 50 > 100 > 200$ (all aligned), APEX registers highest-conviction directional signal.

4.5 SuperTrend (ATR Length: 10, Factor: 3)

- Price above SuperTrend + above 200 EMA = confirmed bullish; only long signals considered
- Price below SuperTrend + below 200 EMA = confirmed bearish; only short signals considered

4.6 Bollinger Bands (20 SMA, 2 SD)

- Band squeeze = imminent volatility expansion; watch for breakout
- Price at upper band + RSI > 70 = overbought warning
- Price at lower band + RSI < 30 = oversold; potential bounce entry

4.7 Pivot Points

Standard Formula:

$$PP = (High + Low + Close) / 3$$

$$R1 = (PP \times 2) - Low \quad | \quad R2 = PP + (High - Low) \quad | \quad R3 = High + 2 \times (PP - Low)$$

$$S1 = (PP \times 2) - High \quad | \quad S2 = PP - (High - Low) \quad | \quad S3 = Low - 2 \times (High - PP)$$

Options strikes near pivot levels define natural stop placement and profit targets.

4.8 Dynamic Support & Resistance

- Identified from high-volume price nodes (volume profile) over the last 52 weeks
- Options strikes near S/R zones flagged as natural anchors for spread legs
- Breakouts above resistance or below support trigger directional signal escalation

SECTION 5 — OPTIONS CHAIN ANALYSIS & GREEKS ENGINE

5.1 Delta-Based Filtering

- **For buying:** Target Delta ≥ 0.50 (high directional leverage; higher ITM probability)
- **For selling:** Target Delta ≤ 0.25 on short legs (high probability of expiring OTM)
- Delta also serves as a probability proxy: 0.25 Delta short put $\approx 25\%$ chance of finishing ITM

5.2 Theta Filtering

- **For buyers:** Low Theta relative to Delta = favorable (option retains value longer per premium dollar)
- **For sellers:** High Theta = favorable (time decay works in seller's favor daily)
- **APEX Delta/Theta Ratio:** Higher ratio = better buying value; lower ratio = better selling value
 - Ratio > 10 : Exceptional buying value
 - Ratio < 3 : Efficient selling candidate

5.3 Vega Assessment

- Vega measures sensitivity to a 1% change in IV
- Pre-catalyst: High Vega = dangerous for buyers if IV crushes post-event
- AI applies a Vega cap rule: long Vega positions in catalyst environments require explicit override or use of Gamma Trampoline structure

5.4 Gamma Awareness

- Gamma measures rate of change in Delta per \$1 move
- Near expiration: Gamma accelerates rapidly (Gamma risk)
- AI flags high-Gamma risk for positions within 7 days of expiration unless structure accounts for it (Gamma Trampoline)

5.5 Bid/Ask Spread Filter

- Hard reject: any option where bid/ask spread exceeds 10% of mid-price
- Wide spreads = hidden cost that destroys edge
- Threshold adjustable in settings (default: 10%; recommended max: 15% for illiquid names)

5.6 Open Interest & Volume Gates

- Minimum OI: 500 contracts per strike
- Volume: must exceed 25% of OI on trade day
- Unusual Options Activity (UOA): volume > 3× average daily → flagged prominently

SECTION 6 — VOLATILITY INTELLIGENCE MODULE

6.1 IV vs. HV Decision Framework

Scenario	Signal	Action
IV << HV (underprice by > 10pts)	Options cheap — BUY premium	Long calls/puts, debit spreads, Gamma Trampoline
IV ≈ HV (within 5pts)	Fair value	Defined-risk spreads, diagonals
IV >> HV (overpriced by > 10pts)	Options rich — SELL premium	Iron condors, credit spreads
IV inversion (front > back)	Catalyst environment	Gamma Trampoline™

6.2 IVR (Implied Volatility Rank) Scoring

- IVR < 25: Cheap IV — buying premium favored
- IVR 25–50: Moderate — systematic income strategies
- IVR 50–75: Rich premium — selling preferred
- IVR > 75: High IV — elevated caution; investigate catalyst

SECTION 7 — SENTIMENT & FUNDAMENTAL LAYER

7.1 Sentiment Analysis Sources

Source	Weight	Signal
Financial News NLP	40%	Earnings surprises, upgrades/downgrades, macro
Options Flow Direction	35%	Net call vs. put dollar flow (48hr window)
Social Media (Reddit/X)	15%	StockTwits + WSJ sentiment, scored -100 to +100
Put/Call Ratio	10%	P/C > 1.2 = fear; P/C < 0.7 = aggressive bullish

7.2 Fundamental Screening

Metric	Signal
EPS Trend	Consecutive beats + revisions upward = bullish boost
Revenue Growth (YoY)	> 15% = strong; < 5% = caution
Earnings Calendar	Days-to-earnings prominently flagged; strategy auto-adjusts
Analyst Consensus	Buy/Hold/Sell ratio + average price target vs. current price
Sector Rotation	Is capital flowing into or out of this sector this week?

SECTION 8 — APEX COMPOSITE SCORE & EXECUTION THRESHOLD

8.1 Scoring Weights

Dimension	Weight
Technical Score (0–100)	30%
Volatility Signal	25%
Greeks Quality Score (0–100)	20%
Sentiment Score (0–100)	15%
Fundamental Score (0–100)	10%
APEX Composite Score	100%

Execution Threshold: Score ≥ 72 / 100 required.

If the score falls below 72, APEX displays:

"Insufficient conviction. Conditions do not meet APEX execution threshold. Monitor for improved setup."

SECTION 9 — THE STRATEGY PLAYBOOK

9.1 Strategy Eligibility Matrix

IV Environment	Direction	Tech Strength	Catalyst	Strategy
IV << HV	Strong Bullish	Score > 80	No	Long Call / Bull Call Spread
IV << HV	Strong Bearish	Score > 80	No	Long Put / Bear Put Spread
IV << HV	Neutral	Medium	No	Long Straddle / Strangle
IV << HV	Any	Any	Yes (5–10 days)	Gamma Trampoline™
IV ≈ HV	Mild Bullish	Medium	No	Diagonal Spread (bullish)

IV Environment	Direction	Tech Strength	Catalyst	Strategy
IV $\approx$ HV	Mild Bearish	Medium	No	Diagonal Spread (bearish)
IV $\approx$ HV	Neutral	Any	No	Calendar Spread
IV $\gg$ HV	Neutral	Low-Medium	No	Short Iron Condor
IV $\gg$ HV	Mild Bullish	Medium	No	Bull Put Spread (credit)
IV $\gg$ HV	Mild Bearish	Medium	No	Bear Call Spread (credit)
IV $\gggg$ HV (extreme)	Any	Any	Immediate	NO TRADE — Wait for IV Crush
Any	Any	Score < 72	Any	NO TRADE — Insufficient Conviction

9.2 APEX Benchmark Greeks Strategy (Proprietary)

Rule 1 — Buy When Delta is High and Theta is Low:

- Condition: Technical + sentiment alignment is bullish/bearish AND IV < HV
- Filter: Delta ≥ 0.55 AND daily Theta < 0.05 per contract
- APEX Delta/Theta Ratio must exceed 10
- Additional gate: bid/ask spread < 8%
- Outcome: Maximum directional leverage with minimal daily time decay cost

Rule 2 — Sell When IV is High, Delta is Low, and Spread is Tight:

- Condition: IVR > 50 AND RSI is neutral (40–60)
- Filter: Short option Delta ≤ 0.20
- Rationale: 80%+ probability of expiring OTM — consistent edge through volume
- Gate: bid/ask spread must be < 10% of mid-price or trade is rejected
- Outcome: Premium collection on high-probability, low-Delta positions

SECTION 10 — THE GAMMA TRAMPOLINE™ (PROPRIETARY STRATEGY)

10.1 Origin & Design Philosophy

The Gamma Trampoline™ is an original strategy developed by the APEX team to solve a specific market problem:

The Problem: Around earnings, IV spikes so aggressively on front-week options that buying a standard straddle becomes prohibitively expensive — yet the move is expected to be large and direction unknown.

The Innovation: Instead of buying both legs at the same expiration (straddle) or the same strike (calendar), the Gamma Trampoline uses **wide OTM strikes across two different expirations simultaneously** — creating a structure that:

1. Subsidizes the cost of the long back-week options by selling front-week volatility at its peak
2. Retains full long-gamma participation after the catalyst event expires
3. Provides meaningful downside protection if the stock moves flat

10.2 The Mechanics

[CATALYST EVENT (e.g., Earnings)]	
Front-Week Expiry (Day After Catalyst)	Back-Week Expiry (Two Weeks Post-Catalyst)
SELL OTM Call [Strike A]	BUY OTM Call [Strike A]
SELL OTM Put [Strike B]	BUY OTM Put [Strike B]
(Same strikes, different expirations – the key innovation)	

Step 1 – Buy the Back-Week Long Options

- Buy 1 OTM Call expiring two weeks after the catalyst at Strike A (above current price)
- Buy 1 OTM Put expiring two weeks after the catalyst at Strike B (below current price)
- These are your long-gamma anchors; they retain substantial time value through and beyond the event

Step 2 – Sell the Front-Week Short Options

- Sell 1 OTM Call expiring the day after the catalyst at the same Strike A
- Sell 1 OTM Put expiring the day after the catalyst at the same Strike B
- These exploit the IV spike that panicked traders create before the event

Net Effect: Premium from front-week shorts heavily subsidizes back-week longs (typically 50–70% offset), resulting in a net cost far below a conventional straddle while maintaining full long-gamma participation through the back week.

10.3 Three Scenarios

Scenario A: Large Post-Earnings Move (The Trampoline Fires)

- Front-week short option is breached but expires the next day with minimal remaining time value
- Back-week long option (now deep ITM with a full week of time value) surges
- The stock "bounces" off the short strike and catapults the long option's profit
- The trade profits because long option value growth outpaces short option loss

Scenario B: Flat Earnings

- Both front-week options expire worthless → full premium collected
- Back-week options now held at dramatically reduced cost

- Full additional week of optionality for exit at breakeven or small profit

Scenario C: Moderate Move

- Front-week options expire with some residual value (manageable loss)
- Back-week options carry both time value and directional component
- Outcome depends on magnitude of move vs. strike distance — AI pre-calculates probability distribution at entry

10.4 Ideal Entry Conditions (AI Trigger Criteria)

Condition	Threshold
Days to Catalyst	5–10 calendar days
Front-Week IVR	> 70 (confirms rich front-week premium)
IV Inversion	Front-week IV must be significantly higher than back-week
Stock Liquidity	ADV > 5M shares; option OI > 1,000 per strike
Stock Profile	Highly liquid tech/growth with documented gap history
Bid/Ask (all 4 legs)	< 8% of mid

10.5 Risk Profile

- **Maximum Loss:** Net debit paid (cost of back-week longs minus front-week premium collected)
- **Maximum Gain:** Theoretically unlimited (call side); substantial (put side)
- **Break-even:** Net cost relative to back-week strike distance
- **Defined-risk:** APEX confirms max loss before presenting the trade card

SECTION 11 — PYTHON TECHNICAL BUILD GUIDE

11.1 Full Python Library & Package List

Below is the complete categorized list of Python packages required to build APEX.

A. Backend API Framework

Package	Version	Purpose
fastapi	0.115+	Async REST API framework — core backend
uvicorn[standard]	0.30+	ASGI server for FastAPI
pydantic	2.7+	Data validation and settings management
pydantic-settings	2.3+	Environment variable configuration

Package	Version	Purpose
python-multipart	0.0.9+	File upload handling
httpx	0.27+	Async HTTP client for external API calls

Install command:

```
pip install fastapi uvicorn[standard] pydantic pydantic-settings python-multipart httpx
```

B. Database & Caching

Package	Version	Purpose
sqlalchemy	2.0+	ORM for PostgreSQL (async support)
asyncpg	0.29+	Async PostgreSQL driver
alembic	1.13+	Database migrations
redis	5.0+	Redis client for real-time caching
aioredis	2.0+	Async Redis operations

Install command:

```
pip install sqlalchemy asyncpg alembic redis aioredis
```

C. Financial Market Data

Package	Version	Purpose
yfinance	0.2.38+	Yahoo Finance: OHLCV, fundamentals, options chain
pandas	2.2+	DataFrame operations for financial data
numpy	1.26+	Numerical computations
tradier	custom	Tradier broker API wrapper (options chain)
alpaca-trade-api	3.0+	Alpaca broker API for execution
ib_insync	0.9.86+	Interactive Brokers API wrapper
polygon-api-client	1.14+	Polygon.io market data API
pytz	2024.1+	Timezone handling for market hours
pandas-market-calendars	4.4+	US market calendar and trading sessions

Install command:

```
pip install yfinance pandas numpy alpaca-trade-api ib_insync polygon-api-client pytz
```

D. Technical Analysis Indicators

Package	Version	Purpose
pandas-ta	0.3.14b	Comprehensive TA library: MACD, RSI, EMA, Bollinger, SuperTrend, Pivots, 130+ indicators
ta	0.11.0	Alternate TA library (backup for specific indicators)
ta-lib	0.4.28+	C-backed TA library for high-performance indicator computation
scipy	1.13+	Statistical calculations (standard deviation, volatility)
statsmodels	0.14+	GARCH models, volatility forecasting

Install command:

```
pip install pandas-ta ta scipy statsmodels
# TA-Lib requires C library install first:
# brew install ta-lib (Mac) or apt-get install libta-lib-dev (Linux)
pip install TA-Lib
```

E. Options Pricing & Greeks Calculation

Package	Version	Purpose
py_vollib	1.0.1	Black-Scholes option pricing: delta, gamma, theta, vega, rho
py_lets_be_rational	1.0.1	Fast rational option pricing (dependency for py_vollib)
mibian	0.1.3	Black-Scholes, Greeks, implied volatility
QuantLib-Python	1.34+	Full quantitative finance library (bond pricing, yield curves)
arch	6.3+	GARCH models for volatility forecasting

Install command:

```
pip install py_vollib py_lets_be_rational mibian arch
# QuantLib requires separate C++ build or use conda:
# conda install -c conda-forge quantlib
```

F. AI & Machine Learning

Package	Version	Purpose
openai	1.35+	GPT-4 API for technical analysis AI model
anthropic	0.28+	Claude API for volatility specialist model

Package	Version	Purpose
google-generativeai	0.7+	Gemini API for sentiment specialist model
langchain	0.2+	Orchestration layer for multi-AI consensus engine
langchain-openai	0.1+	LangChain OpenAI integration
langchain-anthropic	0.1+	LangChain Anthropic integration
transformers	4.41+	HuggingFace models for local NLP
torch	2.3+	PyTorch: neural network backbone for ML models
scikit-learn	1.5+	ML utilities: classification, clustering, feature engineering

Install command:

```
pip install openai anthropic google-generativeai langchain langchain-openai langchain-anthropic
# PyTorch (visit pytorch.org for platform-specific command):
pip install torch torchvision torchaudio
```

G. Sentiment Analysis & NLP

Package	Version	Purpose
finbert-tone	pip install	FinBERT: financial sentiment analysis model
vaderSentiment	3.3.2	Fast lexicon-based sentiment for social media
textblob	0.18+	General NLP and sentiment scoring
newspaper3k	0.2.8	News article scraping and text extraction
feedparser	6.0.11+	RSS feed parsing for financial news
requests	2.32+	HTTP requests for news/social API calls
beautifulsoup4	4.12+	HTML parsing for web scraping
praw	7.7+	Reddit API wrapper (WallStreetBets, options subs)
tweepy	4.14+	Twitter/X API for social sentiment

Install command:

```
pip install vaderSentiment textblob newspaper3k feedparser requests beautifulsoup4 praw
# FinBERT: pip install finbert-tone
```

H. Data Visualization & Charting

Package	Version	Purpose
plotly	5.22+	Interactive charts (served to frontend via API)
matplotlib	3.9+	Static chart generation and export
mplfinance	0.12.10b	Matplotlib extension for candlestick charts
seaborn	0.13+	Statistical visualization for backtesting results

Install command:

```
pip install plotly matplotlib mplfinance seaborn
```

I. Authentication & Security

Package	Version	Purpose
python-jose[cryptography]	3.3.0+	JWT token generation and validation
passlib[bcrypt]	1.7.4+	Password hashing (bcrypt)
pyotp	2.9+	TOTP 2FA implementation
cryptography	42.0+	AES encryption for sensitive data

Install command:

```
pip install python-jose[cryptography] passlib[bcrypt] pyotp cryptography
```

J. Backtesting Engine

Package	Version	Purpose
backtrader	1.9.78+	Python backtesting framework for strategies
zipline-reloaded	3.0+	Quantopian-style strategy backtesting
bt	0.2.10+	Flexible backtesting for portfolio strategies
pyfolio-reloaded	0.9.5+	Performance and risk analytics for backtested portfolios

Install command:

```
pip install backtrader bt pyfolio-reloaded
```

K. Task Queue & Background Jobs

Package	Version	Purpose
celery	5.4+	Distributed task queue for background scans
celery[redis]	5.4+	Redis as Celery message broker
apscheduler	3.10+	Scheduled tasks (market open scan triggers)
flower	2.0+	Celery task monitoring dashboard

Install command:

```
pip install celery[redis] apscheduler flower
```

L. Testing & Code Quality

Package	Version	Purpose
pytest	8.2+	Testing framework
pytest-asyncio	0.23+	Async test support for FastAPI
httpx	0.27+	Test client for FastAPI endpoints
pytest-cov	5.0+	Code coverage reports
black	24.4+	Code formatting
ruff	0.4+	Fast Python linter
mypy	1.10+	Static type checking

Install command:

```
pip install pytest pytest-asyncio pytest-cov black ruff mypy
```

M. Deployment & Infrastructure

Package	Version	Purpose
gunicorn	22.0+	Production WSGI server
python-dotenv	1.0+	Environment variable management
loguru	0.7+	Structured logging
prometheus-fastapi-instrumentator	6.1+	Metrics for Prometheus/Grafana
boto3	1.34+	AWS SDK (S3, EC2, CloudWatch)
sentry-sdk[fastapi]	2.3+	Error tracking and performance monitoring

Install command:

```
pip install gunicorn python-dotenv loguru boto3 sentry-sdk[fastapi] prometheus-fastapi
```

Complete requirements.txt

```
# — Backend Framework —————
fastapi>=0.115.0
uvicorn[standard]>=0.30.0
pydantic>=2.7.0
pydantic-settings>=2.3.0
python-multipart>=0.0.9
httpx>=0.27.0

# — Database & Cache —————
sqlalchemy>=2.0.0
asyncpg>=0.29.0
alembic>=1.13.0
redis>=5.0.0
aioredis>=2.0.0

# — Market Data —————
yfinance>=0.2.38
pandas>=2.2.0
numpy>=1.26.0
alpaca-trade-api>=3.0.0
polygon-api-client>=1.14.0
pytz>=2024.1
pandas-market-calendars>=4.4.0

# — Technical Analysis —————
pandas-ta>=0.3.14b
ta>=0.11.0
scipy>=1.13.0
statsmodels>=0.14.0

# — Options Pricing & Greeks —————
py_vollib>=1.0.1
py_lets_be_rational>=1.0.1
mibian>=0.1.3
arch>=6.3.0

# — AI & Machine Learning —————
openai>=1.35.0
anthropic>=0.28.0
google-generativeai>=0.7.0
langchain>=0.2.0
langchain-openai>=0.1.0
langchain-anthropic>=0.1.0
scikit-learn>=1.5.0

# — Sentiment & NLP —————
vaderSentiment>=3.3.2
```

```

textblob>=0.18.0
newspaper3k>=0.2.8
feedparser>=6.0.11
requests>=2.32.0
beautifulsoup4>=4.12.0
praw>=7.7.0
tweepy>=4.14.0

# — Charting —
plotly>=5.22.0
matplotlib>=3.9.0
mplfinance>=0.12.10b
seaborn>=0.13.0

# — Auth & Security —
python-jose[cryptography]>=3.3.0
passlib[bcrypt]>=1.7.4
pyotp>=2.9.0
cryptography>=42.0.0

# — Backtesting —
backtrader>=1.9.78
bt>=0.2.10

# — Task Queue —
celery[redis]>=5.4.0
apscheduler>=3.10.0
flower>=2.0.0

# — Testing —
pytest>=8.2.0
pytest-asyncio>=0.23.0
pytest-cov>=5.0.0
black>=24.4.0
ruff>=0.4.0
mypy>=1.10.0

# — Deployment —
gunicorn>=22.0.0
python-dotenv>=1.0.0
loguru>=0.7.0
boto3>=1.34.0
sentry-sdk[fastapi]>=2.3.0

```

SECTION 12 — AI MULTI-MODEL CONSENSUS ARCHITECTURE

No trade executes when AI models disagree. Disagreement triggers: *"Hold — Conflicting Signals."*

Model	Role	API
GPT-4 (OpenAI)	Technical Specialist — chart patterns, indicator analysis	openai
Claude (Anthropic)	Volatility Specialist — IV structure, Greeks, options chain	anthropic

Model	Role	API
Gemini (Google)	Sentiment Specialist — news NLP, social scoring	google-generativeai
APEX Arbiter	Composite scoring engine — weights consensus, issues trade brief	Custom logic

Consensus Gate: Minimum 2 of 3 models must agree on direction AND strategy before APEX Composite Score threshold is checked.

SECTION 13 — DEVELOPMENT ROADMAP

Phase 1 — MVP (Months 1–3)

- Ticker search + interactive chart with all 10 indicators
- Live options chain loader with Greeks display
- IV vs. HV calculator + IVR display
- Paper trading mode (no live execution)

Phase 2 — AI Scan Engine (Months 4–6)

- Full 10-dimension scan engine
- Multi-model AI consensus architecture
- Composite APEX Composite Scoring
- Automated strategy recommendation (human approves execution)
- Sentiment NLP integration

Phase 3 — Autonomous Execution (Months 7–9)

- Broker API integration (Interactive Brokers / Alpaca / TD Ameritrade)
- Automated trade execution on APEX Score approval
- Position monitoring + automated exit rules
- Gamma Trampoline™ auto-detection
- Mobile-responsive interface

Phase 4 — Professional Platform (Months 10–12)

- Multi-account management (advisory firms)
- Backtesting engine across all strategies
- Client-facing reporting dashboard
- Regulatory compliance layer (audit trails, position logs)
- Subscription tier management

SECTION 14 — RISK MANAGEMENT FRAMEWORK

- **Position Sizing:** 2–5% of declared account capital per trade (configurable)
- **Portfolio Greeks Monitor:** Running aggregate Delta, Theta, Vega across all open positions
- **Correlation Filter:** Third open position in same sector triggers concentration warning
- **Daily Loss Cap:** Configurable; trading auto-halts if breached
- **Earnings Blackout:** No new positions within 1 day of earnings (exception: Gamma Trampoline setup)
- **Bid/Ask Hard Stop:** No trade executes if any leg spread exceeds 10% of mid-price
- **Vega Cap:** Long Vega positions in catalyst environments require explicit AI override with justification

CLOSING STATEMENT — FOR MENTOR REVIEW

APEX is built on a single conviction: options trading at the professional level should not require a trader to juggle 15 open tabs, three platforms, and manual Greek calculations simultaneously. The analytical infrastructure exists. The AI models exist. The market data exists. What has been missing is a platform that assembles all of it into a single, decisive, explainable, executable workflow.

The Gamma Trampoline™ represents a structural innovation in how traders approach catalyst events — reducing entry cost through IV inversion mechanics while maintaining full long-gamma participation post-event. When embedded within an AI platform that detects precise conditions for its deployment, it becomes a systematic, repeatable edge rather than a one-time trade.

APEX is not trying to replace the trader's judgment — it is designed to give every trader the analytical horsepower of an entire quantitative desk, in under 15 seconds, on any ticker in the market.

We are seeking mentor guidance on: data architecture, AI model integration priority sequencing, regulatory considerations for autonomous execution, and broker API partnership strategy.

This document is confidential and prepared solely for mentor review and advisory purposes. It does not constitute investment advice or a solicitation for investment.

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